

Element C

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Design Requirements:

- At its core, we need a chess board, designed in a 3D CAD software.
 - This is our highest priority requirement, because it is the basic concept of what we are doing.
- Our design must be 3D-printable, and in a file format that can be shared with other people around the world.
 - This is our second highest priority, since if our model is not easily recreated, it will not be able to help as many students.
- Specifically, we want our chess board to have an 8 by 8 square grid, totaling 64 squares.
- There will be 32 pieces (16 for each side). This is composed of a king, queen, 2 bishops, 2 knights, 2 rooks, and 8 pawns.
- Ideally it will be small enough to fit most (>50% of) printers, but also big enough to be used by all kinds of people (e.g. bigger or smaller hands, less dexterity, etc.)
 - Part of this includes being able to easily differentiate between the different types of pieces.
- An optional requirement would be that it has some sort of built-in storage system, to allow students/teachers to bring the game around the school building.
 - If this is not feasible, we could also try to build an external storage system, like a box to hold all the pieces.
 - This is our lowest priority since we want to focus on the actual game-playing experience first, and prioritize storage after this is refined.

Leading to a Solution

- To develop our solution, we will use online 3D modelling software, specifically Onshape's CAD functionality.

- After our solution is created, it will be shared with the world in an easy-to-use file format that can be 3D printed on a wide variety of machines.

Primary Stakeholder Validation

- Our design requirements will help ensure that our final product successfully meets our stakeholders' needs.
- Since our stakeholders are both students and teachers at school, we need to make it accessible to everyone.
 - This is accomplished by our aforementioned design requirements.